

In matrix notation, the single factor model has general form:

$$r_i = \alpha_i + \hat{b}_i R_m + e_i \quad (6.38)$$

r_i = column vector of stock returns for stock i

R_m = column vector of market returns

e_i = column vector of random noise for stock i

$\hat{b}_i$ = stock return sensitivity to market returns

In the single index model we need to estimate the risk exposure $\hat{b}_i$ to the general index R_m . In situations where the index used in the single index model is the broad market index and the constant term is the risk free rate, the single index model is known as the CAPM model (Sharpe, 1964) and the risk exposure $\hat{b}_i$ is the stock beta, β .

Multi-Index Models

The multi-index factor model is an extension of the single index model that captures additional relationships between price returns and corresponding sectors and industries. There have been numerous studies showing that the excess returns (error) from the single index model are correlated across stocks in the same sector, and with further incremental correlation across stocks in the same industry (see Elton and Gruber, 1995).

Let R_m = market returns, S_k = the stock's sector returns, and I_k = the stock's industry return. Then the linear relationship is:

$$r_i = \alpha_i + b_{im}R_m + b_{ik}S_k + b_{il}I_l + e_i$$

where b_{im} is the stock's sensitivity to the general market movement, b_{ik} is the stock's sensitivity to its sector movement, and b_{il} is the stock's sensitivity to its industry movement.

There is a large degree of correlation, however, across the general market, sectors, and industry. These factors are not independent and analysts need to make appropriate adjustment following the process outlined above.

The general multi-index model after multicollinearity now has form:

$$r_i = \alpha_i + \hat{b}_{im}R_m + \hat{b}_{isk}^* \tilde{S}_k + \hat{b}_{ill}^* \tilde{I}_l + \varepsilon \quad (6.39)$$

Macroeconomic Factor Models

A macroeconomic multi-factor model defines a relationship between stock returns and a set of macroeconomic variables such as GDP, inflation, industrial production, bond yields, etc. The appeal of using macroeconomic data as the explanatory factors in the returns model is